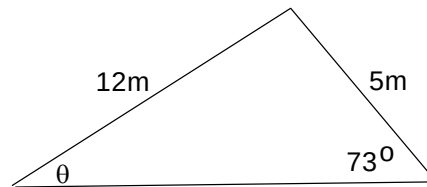


QUESTION 1

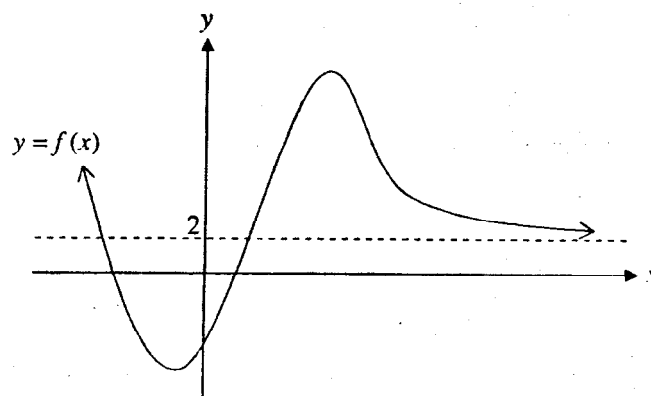
- a) Evaluate $2e^{-0.72}$ correct to 2 significant figures 2
- b) Simplify fully $\sqrt{27} + \sqrt{48} - 2\sqrt{75}$ 2
- c) Solve $|2x + 3| = 3x$ 2
- d) Find the value of θ correct to the nearest degree



- e) Solve $5 - 5x \leq 4$ 2
- f) Find the limiting sum of the geometric series $2 - \frac{2}{3} + \frac{2}{9} - \frac{2}{27} + \dots$ 2

QUESTION 2 (START A NEW PAGE NOW)

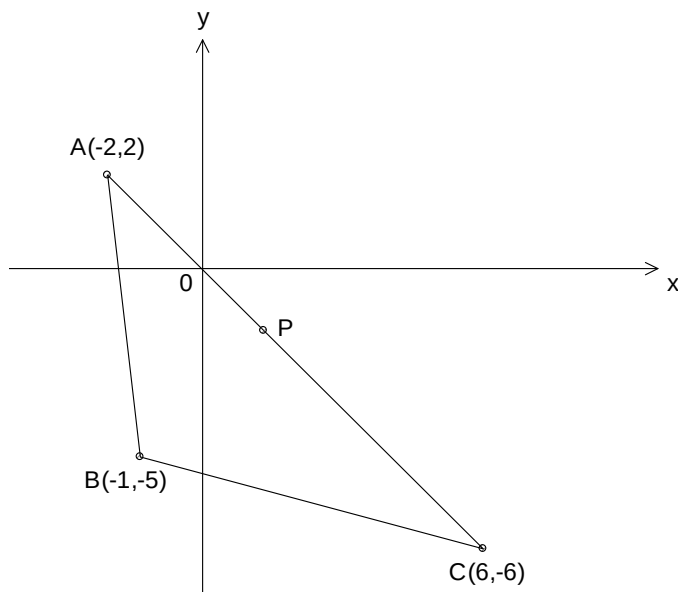
- a) Differentiate with respect to x :
- (i) $x^2 \cdot \cos x$ 2
- (ii) $\tan^2(4x + 1)$ 2
- b) Find $\int \frac{x}{3x^2+1} dx$ 2
- c) Evaluate $\int_0^{\frac{\pi}{9}} \sin 3x \cdot dx$ 3
- d) The diagram below shows a sketch of the curve $y = f(x)$. 3



Copy the diagram onto your answer sheet and use it to draw a sketch of the gradient function $y = f'(x)$

QUESTION 3 (START A NEW PAGE NOW)

- a) The diagram (not to scale) shows $\triangle ABC$ with vertices $A(-2,2)$, $B(-1,-5)$ and $C(6,-6)$



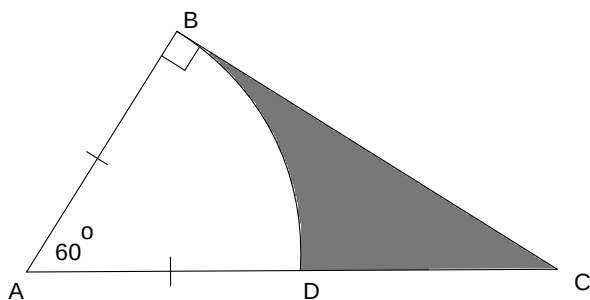
Copy the diagram neatly onto your answer sheet

- | | |
|---|---|
| (i) P is the midpoint of AC. Show that the coordinates of P are (2,-2). | 1 |
| (ii) Find the gradient of BP. | 1 |
| (iii) Show that BP is perpendicular to AC. | 2 |
| (iv) Show that the equation of BP is $y = x - 4$. | 1 |
| (v) Find the coordinates of D, if P is the midpoint of BD. | 2 |
| (vi) Which shape best describes the geometric figure ABCD? Explain. | 2 |

- b) Find the value(s) of k such that the roots of $x^2 + (3k + 1)x + 4k + 5 = 0$ are real and equal. 3

QUESTION 4 (START A NEW PAGE NOW)

- a) In the diagram, $\angle B = 90^\circ$, $\angle A = 60^\circ$ and $AB = AD = 10\text{m}$. BD is an arc of the circle with centre A



Calculate the shaded area in exact form.

3

- b) In an arithmetic progression, $T_2 = 7$ and $T_7 = 52$.
- Find the common difference and the first term. 2
 - Find the **value** of the first term which is greater than 1000. 2
- c) α and β are the roots of $2x^2 - 9x - 3 = 0$. Find the value of:
- $\alpha + \beta$ 1
 - $\alpha\beta$ 1
 - $\alpha^2 + \beta^2$ 1
 - $\alpha^3 + \beta^3$ 2

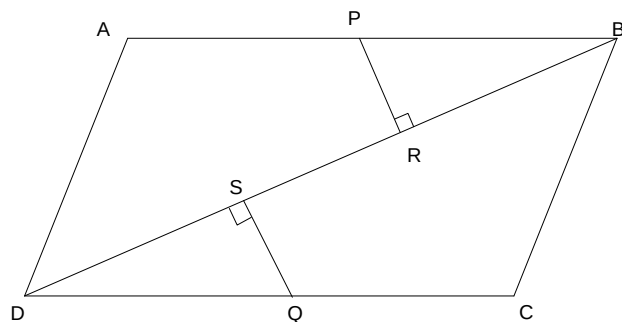
QUESTION 5 (START A NEW PAGE NOW)

- Find the coordinates of the point on the curve $y = 3x^2 - 2x - 1$ where the tangent is parallel to $4x - y - 1 = 0$. 3
- Solve $\sin x = \frac{1}{2}$ where $0 < x < 2\pi$ 2
- Evaluate 1

$$\sum_{n=4}^7 \frac{1}{n-2}$$
- The equation $(x - 1)^2 = -4y + 2$ represents a parabola.
 - Find the coordinates of the vertex. 1
 - Find the focal length. 1
 - Sketch the parabola, clearly showing the directrix, the focus and the x-intercepts 4

QUESTION 6 (START A NEW PAGE NOW)

- ABCD is a parallelogram and P, Q are the midpoints of AB, DC respectively.
 The intervals PR and QS are perpendicular to the diagonal DB.

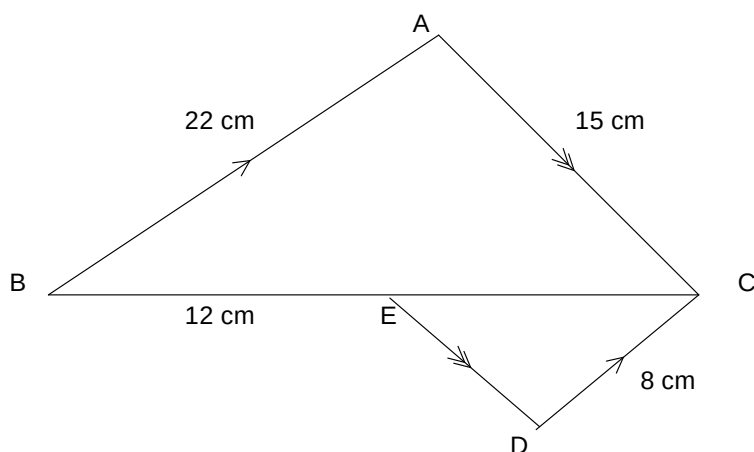


- Prove $\triangle BPR$ and $\triangle DQS$ are congruent. 3
- If $AB=10\text{cm}$, $PR=3\text{cm}$ and $BD=14\text{cm}$ find the length of SR. 1

- b) (i) Sketch the graph of $y = 2 \cos 2x$ for $-\pi \leq x \leq \pi$. 2
- (ii) On the same diagram, sketch the line $x + y = 1$. 1
- (iii) Hence determine the **number** of solutions of the equation $2 \cos 2x = 1 - x$. 1
- (iv) Let the negative solution to $2 \cos 2x = 1 - x$ be $x = N$. Indicate N on the x -axis of your diagram. 1
- c) Calculate the exact area of the region bounded by the curve $y = e^{2x}$, the y -axis and the line $y = e^4$. 3

QUESTION 7 (START A NEW PAGE NOW)

- a) A circle has radius 2 cm. Find the size of the angle subtended at the centre of this circle by an arc of length 2.2 cm. Answer correct to the nearest minute . 2
- b) Solve $\log_e x - \log_e (x - 4) = \log_e 2$ 2
- c) Find all values of k for which $y = e^{kx}$ is a solution of $y'' - y' - 12y \geq 0$ 4
- d) In the diagram (not to scale), CD is parallel to AB and DE is parallel to CA. AC=15 cm, AB=22 cm, CD=8 cm and BE=12 cm. 2

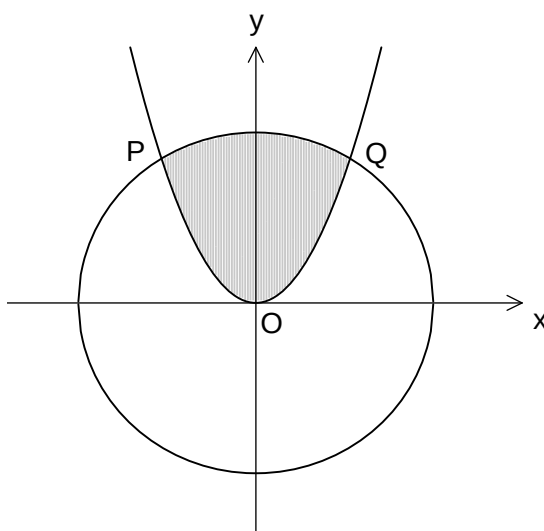


- (i) Prove triangle ABC is similar to triangle DCE 2
- (ii) Hence find the length of BC 2

QUESTION 8 (START A NEW PAGE NOW)

- a) (i) Show that $\int_0^2 \frac{1}{1+x} dx = \ln 3$ **1**
- (i) Hence use Simpson's rule with three function values to find an approximation to $\ln 3$. **2**

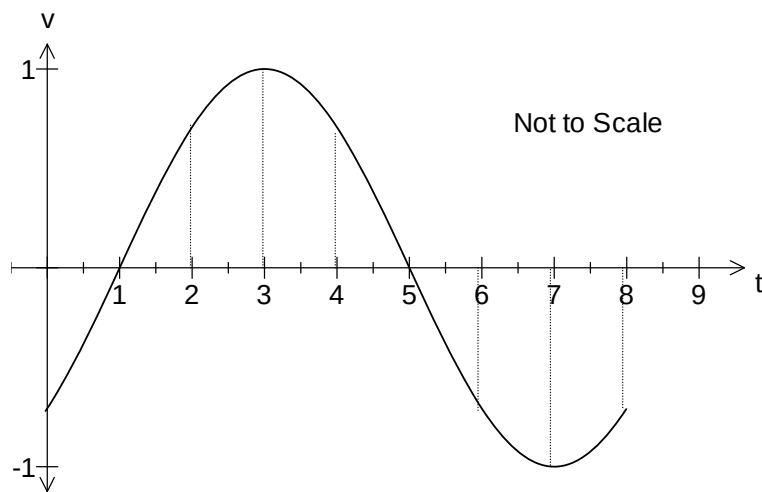
- b) The region bounded by the parabola $y = x^2$ and the circle $x^2 + y^2 = 12$ is shown. The parabola and circle intersect at P and Q.



- (i) P and Q have the same y-coordinate. Find its value. **1**
- (ii) Find the exact value of the solid generated when the shaded region is rotated about the y-axis. **5**
- c) Kim and Lee toss a biased coin alternately, with Kim going first. The probability that the coin shows 'heads' on any toss is $\frac{1}{3}$. The first person to throw a head wins the game. What is the probability that:
- (i) Kim wins the game on her first throw? **1**
- (ii) Lee wins on his first throw? **1**
- (iii) after 4 tosses of the coin, there is no winner? **1**

QUESTION 9 (START A NEW PAGE NOW)

- a) The diagram shows the velocity-time graph for a particle moving in a straight line.



State the times between $t=0$ and $t=8$ at which:

- | | |
|-------------------------------|----------|
| (i) The velocity is zero | 1 |
| (ii) The acceleration is zero | 1 |
| (iii) The speed is increasing | 2 |

- b) The population of Town a is $P_A = 8000e^{0.02t}$ while the population of Town B is $P_B = 12000e^{-0.01t}$, where t is the time in years. How many years will it be until the two populations are equal? Answer correct to 1 decimal place. **2**

- c) When Susan was born, her father deposited \$200 into a Trust account earning 12% p.a. interest compounding annually. The interest is paid on her birthday each year. Susan's father decided to deposit an additional \$200 into this account on her birthday each year, immediately after the interest is received.

- (i) Find the value that the initial deposit would amount to on her 18th birthday. **1**

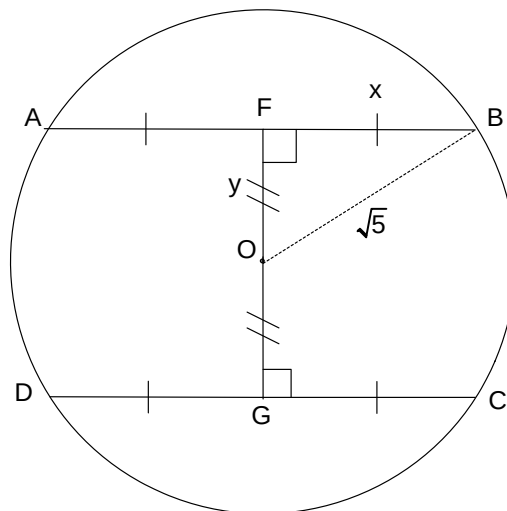
- (ii) Let A_n = total amount in her account on her n th birthday
 Show that $A_n = \frac{5000(1.12^{n+1}-1)}{3}$ **3**

- (iii) Susan's father made his last deposit of \$200 on her 17th birthday. On her eighteenth birthday, he gave all the money in the account to Susan. How much money did she receive? **2**

QUESTION 10 (START A NEW PAGE NOW)

- a) The gradient function of a curve $y = f(x)$ is given by $f'(x) = x^2(3 - x)$.
- (i) Show that the curve $y = f(x)$ has two stationary points and determine their nature. 4
- (ii) If $f(0) = 2$ and the maximum value of $f(x)$ is $8\frac{3}{4}$, draw a possible sketch of $y = f(x)$ 2

- b) A circular window of radius $\sqrt{5}$ metres requires three metal strips AB, DC and FG for reinforcement as shown. O is the centre of the window, and $OF = OG = y$ metres and $FB = FA = CG = GD = x$ metres.



- (i) Given that L is the total length of the metal strips (ie. $L = AB + CD + FG$) show that $L = 4x + 2\sqrt{5 - x^2}$ 1
- (ii) The window will have maximum strength when the total length L is a maximum. Find the value of x for which the window has maximum strength. 5

End of examination

QUESTION 1 Solutions: 2 unit Trial 2010

a) 0.9735045

= 0.97 (2sf)

(1)

(1)

Total /120

b) $3\sqrt{3} + 4\sqrt{3} - 10\sqrt{3}$

= $-3\sqrt{3}$

(2)

-1 any error

c) $2x+3 = 3x$ or

$2x+3 = -3x$

$3 = x$

$5x = -3$

Test

LHS = $|6+3| = 9$ ✓

RHS = $3 \times 3 = 9$

$x = -3/5$

Test

LHS = $| -6/5 + 3 | = 9/5$

RHS = $-9/5$

(1) both solns

not equal

$\therefore x = 3$ only.

(1) final one soln only.

d) $\frac{5}{\sin \theta} = \frac{12}{\sin 73^\circ}$

(1)

$12 \sin \theta = \frac{5 \sin 73^\circ}{12} = 0.3985$

$\theta = 23^\circ$

(1)

e) $-5x \leq -1$

(1)

$x \geq 1/5$

(1)

f) $a=2, r=-1/3$

(1)

$S_\infty = \frac{a}{1-r} = \frac{2}{1-(-1/3)}$

$= \frac{2}{4/3}$

$= 1.5$ (1)

QUESTION 2.

a) i) $y = x^2 \cdot \cos x$

$y' = uv' + vu'$ $\left[\begin{array}{l} u = x^2 \\ v = \cos x \end{array} \right. \quad \left. \begin{array}{l} u' = 2x \\ v' = -\sin x \end{array} \right]$

$= x^2 \cdot (-\sin x) + \cos x \cdot 2x$
 $= -x^2 \sin x + 2x \cos x$

ii) $y = \tan^2(4x+1)$

$y' = 2 \tan(4x+1) \cdot \sec^2(4x+1) \cdot 4$

$= 8 \tan(4x+1) \cdot \sec^2(4x+1)$ (2)

b) $\int \frac{x}{3x^2+1} dx$

$= \frac{1}{6} \int \frac{6x}{3x^2+1} dx$ $\frac{f'(x)}{f(x)}$

$= \frac{1}{6} \ln(3x^2+1) + c$

(1) for $\ln(3x^2+1)$
 (2) for $1/6$

c) $\int_0^{\pi/9} \sin 3x \cdot dx$

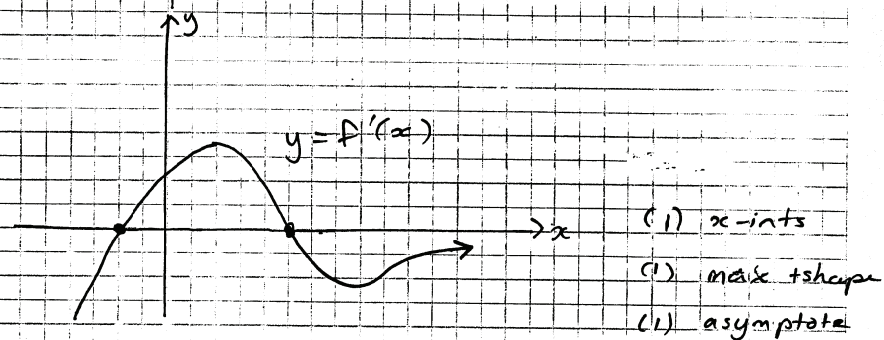
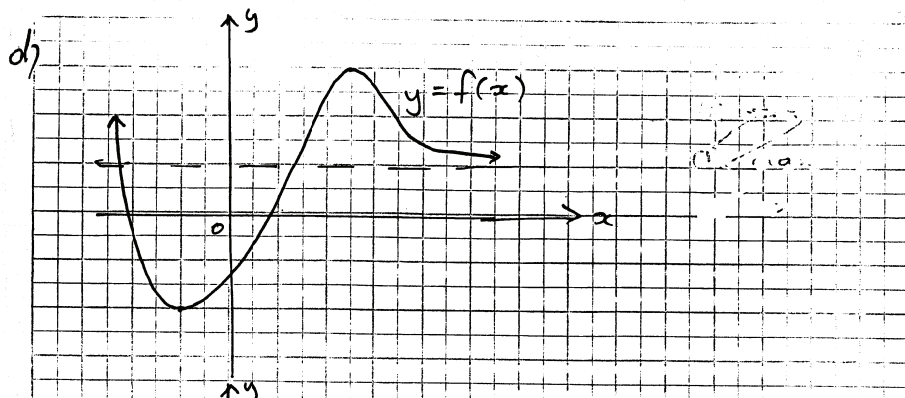
$= -\frac{1}{3} [\cos 3x]_0^{\pi/9}$ (2) (1)

$= -\frac{1}{3} (\cos \frac{\pi}{3} - \cos 0)$

$= -\frac{1}{3} (\frac{1}{2} - 1)$

$= -\frac{1}{3} \times -\frac{1}{2}$

$= \frac{1}{6}$ (1)



QUESTION 3

a) i) $P = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$
 $= \left(\frac{-2+6}{2}, \frac{2-6}{2} \right)$ (1)
 $= \left(\frac{4}{2}, \frac{-4}{2} \right) = (2, -2)$

ii) $m_{BP} = \frac{y_2 - y_1}{x_2 - x_1}$ $B(-1, -5)$
 $P(2, -2)$
 $= \frac{-2 - (-5)}{2 - (-1)}$
 $= \frac{-2+5}{2+1}$
 $= \frac{3}{3} = 1$ (1)

iii) $m_{AC} = \frac{6-2}{6-(-2)} = \frac{-8}{8} = -1$ (1)

$\therefore m_{BP} \times m_{AC} = 1 \times -1 = -1 \therefore \text{Perp.}$ (1)

iv) BP has $m=1$ and thr. $(-1, -5)$
 $y+5 = 1(x+1)$ (1)
 $y+5 = x+1$
 $y = x-4$

v) $\frac{x-1}{2} = 2$ $\frac{y-5}{2} = -2$
 $x-1 = 4$ $y-5 = -4$
 $x = 5$ $y = 1$

$\therefore D(5, 1)$ (1) for x
 (1) for y

vi) Diagonals bisect and are perpendicular
 $\therefore$ Rhombus $\leftarrow$ (1)

b) $\Delta = (3k+1)^2 - 4 \cdot 1 \cdot (4k+5)$
 $= 9k^2 + 6k + 1 - 16k - 20$
 $= 9k^2 - 10k - 19$ (1)
 $= 0$ for real + equal roots (1)
 $k = \frac{10 \pm \sqrt{(-10)^2 - 4 \times 9 \times -19}}{2 \times 9}$
 $= \frac{10 \pm \sqrt{784}}{18}$
 $= \frac{38}{18}, -1 = \frac{19}{9}, -1$ (1)

QUESTION 4

a) Area shaded

$$= \frac{1}{2} \cdot \frac{10}{\sqrt{3}} \cdot BC - \frac{1}{6} \cdot \pi \cdot 10^2$$

$$= 5 \cdot BC - \frac{100\pi}{6} \quad (1)$$

Now $\tan 60^\circ = \frac{BC}{10}$, $BC = 10 \tan 60^\circ$
 $= 10\sqrt{3} \quad (1)$

$$\therefore \text{Area} = 5 \times 10\sqrt{3} - \frac{100\pi}{6}$$

$$= 50\sqrt{3} - \frac{50\pi}{3} \quad \text{units}^2 \quad (1)$$

b) i) $T_2 = a + d = 7$

$T_7 = a + 6d = 52$

$5d = 45$

$d = 9 \quad (1)$

$a + 9 = 7$

$a = -2 \quad (1)$

ii) $T_n = a + (n-1)d$

$= -2 + (n-1) \cdot 9$

$= -2 + 9n - 9$

$= 9n - 11 > 1000 \quad (1)$

when $9n > 1011$

$n > 112.3$

$\therefore T_{113} = 9(113) - 11 = 1006 \quad (1)$

c) $2x^2 - 9x - 3 = 0$

(i) $\alpha + \beta = \frac{9}{2} \quad (1)$

(ii) $\alpha\beta = -\frac{3}{2} \quad (1)$

(iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$= \left(\frac{9}{2}\right)^2 - 2\left(-\frac{3}{2}\right)$

$= 23\frac{1}{4} \quad \text{or} \quad \frac{93}{4} \quad (1)$

(iv) $\alpha^3 + \beta^3 = (\alpha + \beta)(\alpha^2 - \alpha\beta + \beta^2) \quad (1)$

$= \frac{9}{2} \times \left(23\frac{1}{4} - \left(-\frac{3}{2}\right)\right)$

$= 111.375 \quad \text{or} \quad \frac{891}{8} \quad (1)$

QUESTION 5

a) $m = \frac{dy}{dx} = 6x - 2$

Line has $m = 4$

$\therefore 6x - 2 = 4$

$6x = 6$

$x = 1 \quad (1)$

$y = 3(1)^2 - 2(1) - 1 = 0$

$\therefore$ At point $(1, 0) \quad (1)$

b) $x = \frac{\pi}{6}, \frac{5\pi}{6} \quad (2) \quad [1 \text{ each}]$

-1 for degrees.

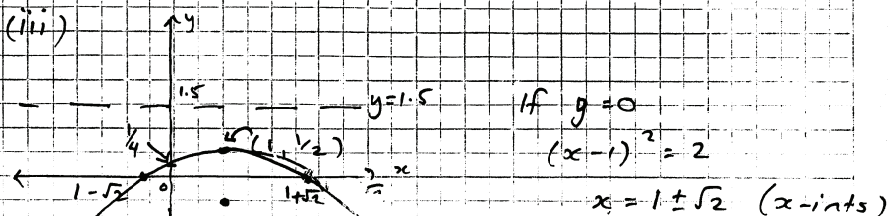
c) $\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} = 1\frac{17}{60} \quad \text{or} \quad \frac{77}{60} \quad (1)$

d) $(x-1)^2 = -4(y-\frac{1}{2})$

(i) Vertex $(1, \frac{1}{2})$ (1)

(ii) $a = 1$ (1)

(iii)



If $y = 0$

$$(x-1)^2 = 2$$

$$x = 1 \pm \sqrt{2} \quad (x\text{-ints})$$

If $x = 0$

(i) Conc. down, vertex
(i) Dir.
(i) Focus
(i) x-int (one pos, one neg)
Not needed.

$$y = 1 = -4y + 2$$

$$-1 = -4y$$

$$y = \frac{1}{4} \quad (y\text{-int})$$

QUESTION 6

a) i) In $\triangle BPR$, $\triangle DQS$:

$$\angle PRB = \angle DSQ \quad (\text{given, } 90^\circ \text{ each})$$

$AB \parallel DC$ (opp sides of parm. $\parallel$)

$$\angle PBR = \angle SDQ \quad (\text{alternate } \angle s, AB \parallel DC) \quad (1)$$

$$BP = \frac{1}{2} \cdot AB$$

$$DQ = \frac{1}{2} \cdot DC \quad (P, Q \text{ are midpts of sides})$$

but $AB = DC$ (opp sides of parm equal)

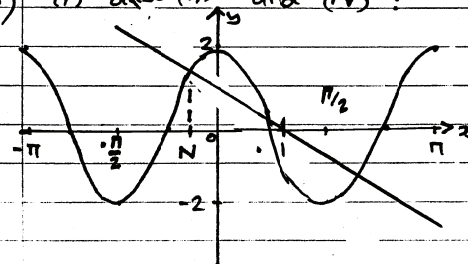
$$\therefore BP = DQ \quad (\text{halving equal sides}) \quad (1)$$

$$\therefore \triangle BPR \equiv \triangle DQS \quad \text{by AAS.} \quad (1)$$

(ii) $14 - 4 - 4 = 6 \text{ cm}$

(1)

b) (i) and (ii) and (iv):



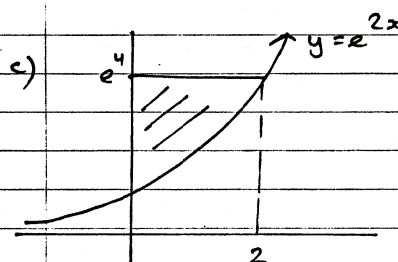
(i) $Pd = 2\pi$ (1)

$Amp = 2$ (1)

(ii) Line (1)

(iii) 3 (1)

(iv) Show N (1)



Area under curve

$$= \int_0^2 e^{2x} \cdot dx$$

$$= \left[\frac{1}{2} e^{2x} \right]_0^2 \quad (1)$$

$$= \frac{1}{2} e^4 - \frac{1}{2} e^0$$

$$= \frac{1}{2} e^4 - \frac{1}{2} \quad (1)$$

$$\text{Area shaded} = 2e^4 - \left(\frac{1}{2} e^4 - \frac{1}{2} \right) \quad (1)$$

$$= \frac{3}{2} e^4 + \frac{1}{2} \text{ units}^2$$

QUESTION 7

a) $l = c\theta$ $2.2 = 2\theta$

$$\theta = 1.1 \text{ radians} \quad (1)$$

$$= 1.1 \times \frac{180}{\pi} \text{ deg.}$$

$$= 63^\circ 2'$$

must be nearest (1)

$$b) \log_e \frac{x}{x-4} = \log_e 2 \quad (1)$$

$$\frac{x}{x-4} = 2$$

$$x = 2x - 8$$

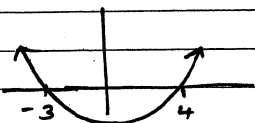
$$\underline{8 = x} \quad (1)$$

$$c) \left. \begin{array}{l} y = e^{kx} \\ y' = ke^{kx} \quad (1) \\ y'' = k^2 e^{kx} \quad (1) \end{array} \right\} \begin{array}{l} k^2 e^{kx} - ke^{kx} - 12e^{kx} \geq 0 \\ e^{kx} (k^2 - k - 12) \geq 0 \\ [\div e^{kx} \text{ and factorise}] \quad (k-4)(k+3) \geq 0 \quad (1) \end{array}$$

$$\text{Roots: } k = 4, -3$$

Soln:

$$k \geq 4, \quad k \leq -3 \quad (1)$$



$$d) (i) \left. \begin{array}{l} \angle DCE = \angle ABC \quad (\text{alternate } \angle s, AB \parallel DC) \\ \angle CED = \angle ACB \quad (\text{alternate } \angle s, AC \parallel DE) \end{array} \right\} (1)$$

$$\therefore \triangle ABC \parallel \triangle DCE \quad (\text{matching } \angle s \text{ equal}) \quad (1)$$

$$(ii) \text{ Let } EC = x \quad \frac{x}{x+12} = \frac{8}{22} \quad (1)$$

$$11x = 4x + 48$$

$$7x = 48$$

$$x = \frac{48}{7}$$

$$\therefore BC = 12 + \frac{48}{7} = \frac{132}{7} \quad (1)$$

QUESTION 8

$$a) (i) \int_0^2 \frac{1}{1+x} \cdot dx = [\ln(1+x)]_0^2$$

$$= \ln 3 - \ln 1 \quad (1)$$

$$= \ln 3$$

$$(ii) \begin{array}{c|ccc} x & 0 & 1 & 2 \\ \hline y = \frac{1}{x+1} & 1 & \frac{1}{2} & \frac{1}{3} \end{array} \quad (1)$$

$$\ln 3 \div \frac{1}{3} (y_0 + 4y_1 + y_2)$$

$$= \frac{1}{3} (1 + 4 \times \frac{1}{2} + \frac{1}{3}) \quad (1)$$

$$= \frac{10}{9}$$

$$b) (i) y = x^2, \quad x^2 + y^2 = 12$$

$$y + y^2 = 12$$

$$y^2 + y - 12 = 0$$

$$(y+4)(y-3) = 0$$

$$\underline{y = -4} \quad \text{or} \quad \underline{y = 3} \quad (1)$$

$$(ii) V_1 = \pi \int_0^3 y \cdot dy$$

$$= \pi \left[\frac{y^2}{2} \right]_0^3$$

$$= \frac{9\pi}{2} \quad (1)$$

$$V_2 = \pi \int_3^{\sqrt{12}} (12 - y^2) \cdot dy \quad \text{Limits } 3, \sqrt{12} : (1)$$

$$= \pi \left[12y - \frac{y^3}{3} \right]_3^{\sqrt{12}} \quad (1)$$

$$= \pi \left((12\sqrt{12} - \frac{12\sqrt{12}}{3}) - (36 - 9) \right)$$

$$= \pi (8\sqrt{12} - 27)$$

$$= \pi (16\sqrt{3} - 27) \quad (1)$$

$$\therefore \text{Total vol} = V_1 + V_2$$

$$= \frac{9\pi}{2} + \pi (16\sqrt{3} - 27) \quad (1)$$

$$= \pi \left(\frac{9}{2} + 16\sqrt{3} - 27 \right)$$

$$= \pi (16\sqrt{3} - \frac{45}{2})$$

$$c) (i) \frac{1}{3} \quad (1)$$

$$(ii) \frac{2}{3} \times \frac{1}{3} = \frac{2}{9} \quad (1)$$

$$(iii) \left(\frac{2}{3}\right)^4 = \frac{16}{81} \quad (1)$$

QUESTION 9.

$$a) (i) t = 1, 5 \quad (1)$$

$$(ii) t = 3, 7 \quad (1)$$

$$(iii) \left. \begin{array}{l} v \text{ pos} \\ a \text{ pos} \end{array} \right\} \therefore 1 < t < 3 \quad (1)$$

or

$$\left. \begin{array}{l} v \text{ neg} \\ a \text{ neg} \end{array} \right\} \therefore 5 < t < 7 \quad (1)$$

$$b) 8000e^{0.02t} = 12000e^{-0.02t}$$

$$e^{0.03t} = 1.5 \quad (1)$$

$$0.03t = \ln 1.5$$

$$t = \frac{\ln 1.5}{0.03} = 13.5 \text{ yrs} \quad (1)$$

$$c) (i) A = 200(1.12)^{18} = \$1537.99 \quad (1)$$

$$(ii) A_1 = 200(1.12) + 200$$

$$A_2 = A_1 \times 1.12 + 200$$

$$= 200(1.12)^2 + 200(1.12) + 200 \quad \text{Developme}$$

$$A_3 = A_2 \times 1.12 + 200 \quad (1)$$

$$= 200(1.12)^3 + 200(1.12)^2 + 200(1.12) + 200$$

$$= 200(1.12^3 + 1.12^2 + 1.12 + 1)$$

Continuing the pattern:

$$A_n = 200(1.12^n + 1.12^{n-1} + \dots + 1.12 + 1) \quad (1)$$

$$= 200 \times \frac{1(1.12^{n+1} - 1)}{1.12 - 1} \quad \begin{array}{l} \text{geom series} \\ a=1 \\ r=1.12 \end{array}$$

$$= 200 \times \frac{(1.12^{n+1} - 1)}{0.12} \quad (1) \quad \begin{array}{l} (n+1) \text{ term} \\ S_n = \frac{a(r^n - 1)}{r - 1} \end{array}$$

$$= \frac{5000(1.12^{n+1} - 1)}{3}$$

$$(iii) \text{Final amt} = A_{17} \times 1.12 \quad \text{or} \quad A_{18} - 200$$

$$= \$11150 \times 1.12 \quad \text{or} \quad \$12688 - 200$$

$$= \$12488 \quad (2)$$

-1 if out by 200

QUESTION 10

a) (i) Start pts when $f'(x) = x^2(3-x) = 0$ (1)
 i.e. When $x=0$ or $x=3$ (1)

Testing

x	-1	0	1
$f'(x)$	+	0	+

$\therefore$ Horizontal pt of inflex when $x=0$. (1)

$$(-1)^2 \cdot (3+1) = 1 \times 4$$

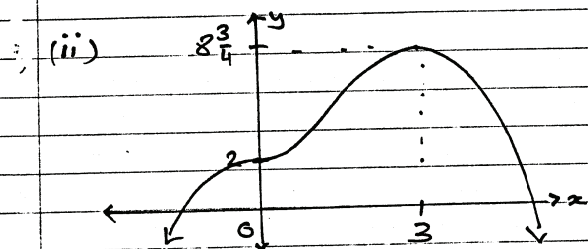
$$1^2(3-1) = 1 \times 2$$

x	2	3	4
$f'(x)$	+	0	-

$\therefore$ Max turn pt at $x=3$ (1)

$$2^2(3-1) = 4 \times 1$$

$$4^2(3-4) = 16 \times -1$$



(2)

b) (i) $L = AB + CD + FG$
 $= 2x + 2x + 2y$ but $\begin{cases} x^2 + y^2 = 5 \\ y = \sqrt{5-x^2} \end{cases}$
 $= 4x + 2\sqrt{5-x^2}$ (1)

(ii) For max L , $L' = 0$ and $L'' < 0$

$$L' = 4 + \cancel{2} \cdot \frac{1}{2} (5-x^2)^{-1/2} \cdot -2x = 4 - \frac{2x}{\sqrt{5-x^2}} = 0 \text{ when } \dots$$
 (1)

$$= 4 - \frac{2x}{\sqrt{5-x^2}} = 0 \text{ when } \dots$$

$$\frac{2x}{\sqrt{5-x^2}} = 4 \quad (1)$$

$$2x = 4\sqrt{5-x^2}$$

Square both sides:

$$x^2 = 4(5-x^2)$$

$$x^2 = 20 - 4x^2$$

$$5x^2 = 20$$

$$x^2 = 4$$

$$x = \pm 2$$

but $x > 0$

$$x = 2$$

(1)

Testing with L'' (since L'' is messy)
 $\therefore$ [Can't test with $x=$

x	1	2	2.1
L'	+	0	-

$$4 - \frac{2}{\sqrt{4}}$$

$$4 - \frac{4.2}{\sqrt{5-4.41}}$$

(2)

Deduct 1 if uncle

$\therefore L$ is a max when $x=2$.

If they use L'' (not recommended)

$$L' = 4 - \underbrace{2x(5-x^2)^{-1/2}}_{\substack{u=2x \quad u'=2 \\ v=(5-x^2)^{-1/2} \quad v'=-\frac{1}{2}(5-x^2)^{-3/2} \\ = x(5-x^2)^{-3/2} \\ \text{(product rule)}}}$$

$$L'' = 0 - \left(\frac{2x \cdot x(5-x^2)^{-3/2} + (5-x^2)^{-1/2} \cdot 2}{(5-x^2)^{-3/2}} \right)$$

$$= - \frac{2x}{(5-x^2)^{3/2}} - \frac{2}{\sqrt{5-x^2}} \quad (1)$$

$$= \frac{-4}{1^{3/2}} - \frac{2}{\sqrt{5-4}} \quad \text{when } x=2$$

$$= -4 - 2 \quad (1)$$

$$= -6 < 0 \quad \underline{\text{Max } L \text{ when } x=2}$$

OR

$$L'' = 4 - \left[\frac{2x}{(5-x^2)^{1/2}} \right] \quad \substack{u=2x \quad u'=2 \\ v=(5-x^2)^{1/2} \quad v'=\frac{1}{2}(5-x^2)^{-1/2} \\ = -\frac{x}{\sqrt{5-x^2}}}$$

$$= - \left(\frac{\sqrt{5-x^2} \cdot 2 - 2x \cdot \frac{-x}{\sqrt{5-x^2}}}{5-x^2} \right) \quad (1)$$

$$= - \left(\frac{2\sqrt{5-x^2} + \frac{2x^2}{\sqrt{5-x^2}}}{5-x^2} \right)$$

$$= - \left(\frac{2\sqrt{1} + \frac{8}{\sqrt{1}}}{1} \right) \text{ when } x=2 \quad (1) \quad = -(2+8) < 0 \therefore \text{Max}$$